Summer Examinations 2020/2021

Course Instance Codes	2BDS1, 3BA1, 3BS9, 3BPT1, 4BS2, 4BME1, 1CSD1, 1CSD2
Exams	Second Arts with Data Science, Third Arts, Third Science, Third Bachelor of Science (Physics), Fourth Science, Fourth Mathematics and Education, MSc Data Analytics
Module Codes	**ST312**
Module	Applied Statistics II
External Examiner	Professor S. McClean
Internal Examiner(s)	Dr. A. Simpkin Prof. J. Newell
Instructions:	**Answer ALL FIVE questions** **This is an OPEN BOOK exam** **Issues can be emailed to andrew.simpkin@nuigalway.ie**
Integrity:	**In submitting this work you confirm that it is entirely your own. You acknowledge that you may be invited to online interview if there is any concern in relation to the integrity of your exam.**
Duration	2 Hours
No. of Pages	14 pages, including this one
School	School of Mathematics, Statistics and Applied Mathematics
Release in Exam Venue	Yes ☑ No ☐

Requirements:

Statistical Tables/ Log Tables	Relevant distribution tables are attached to this paper. The *New Formulae & Tables* are optional.

Question 1: [Total marks 15]

To determine how a new type of short wave UVA-blocking sunscreen affects the general health of skin in comparison to a regular long wave UVA-blocking sunscreen, a random sample of 40 participants were randomly separated into equal groups of 20: an experimental group and a control group. All participants' skin health was then initially evaluated. The experimental group wore the short wave UVA-blocking sunscreen daily, and the control group wore the long wave UVA-blocking sunscreen daily. After one month, the general health of the skin was measured in both groups and statistically analyzed.

Figure 1:

a) Is this an observational or experimental study? Explain your answer.
b) Identify the response and explanatory variables, the treatments, and the experimental units.
c) Are the results generalisable from this sample to the population of all individuals? Why/why not? (2 marks)
d) Why were participants randomly assigned to groups? Explain the advantages of this approach. (3 marks)
e) Does this experiment give a fair comparison between the two treatments? Why? (2 marks)
f) How could this study design be improved? (2 marks)

Question 2: [Total marks 15]

A farmer would like to compare three different types of corn to see which produces the highest yield (in kilograms). The farmer has ten fields to use as part of this study to find out if the types have the same yield. In accordance with the randomized block design, each type of corn will be planted in subplots within all ten fields. The planting location of corn types in each field was randomly assigned.

Each row in the following table represents a field, while the columns show the three corn types (A,B,C) and the values show the yield of corn.

Field	Corn A	Corn B	Corn C
1	378	419	289
2	310	376	193
3	453	529	406
4	218	318	148
5	427	536	406
6	329	417	340
7	425	529	346
8	291	318	248
9	412	436	346
10	392	417	340

a) List the factor of interest and any nuisance factors. (2 marks)
b) What are the experimental units? (1 mark)
c) Are there blocks in this design? If so, identify them. (2 marks)
d) Why did the farmer conduct the experiment in this way? (2 marks)
e) Has their approach been justified by the data? (2 marks)
f) Describe how a randomisation test could be used to see if the items differ in mean yield, taking account of the possible between *field* variation. State the null and alternative hypotheses being tested and describe (briefly) how such a test would be implemented including how the associated p-value would be calculated. What would be the advantage of doing this rather than a standard ANOVA F-test? (No calculations are required.) (6 marks)

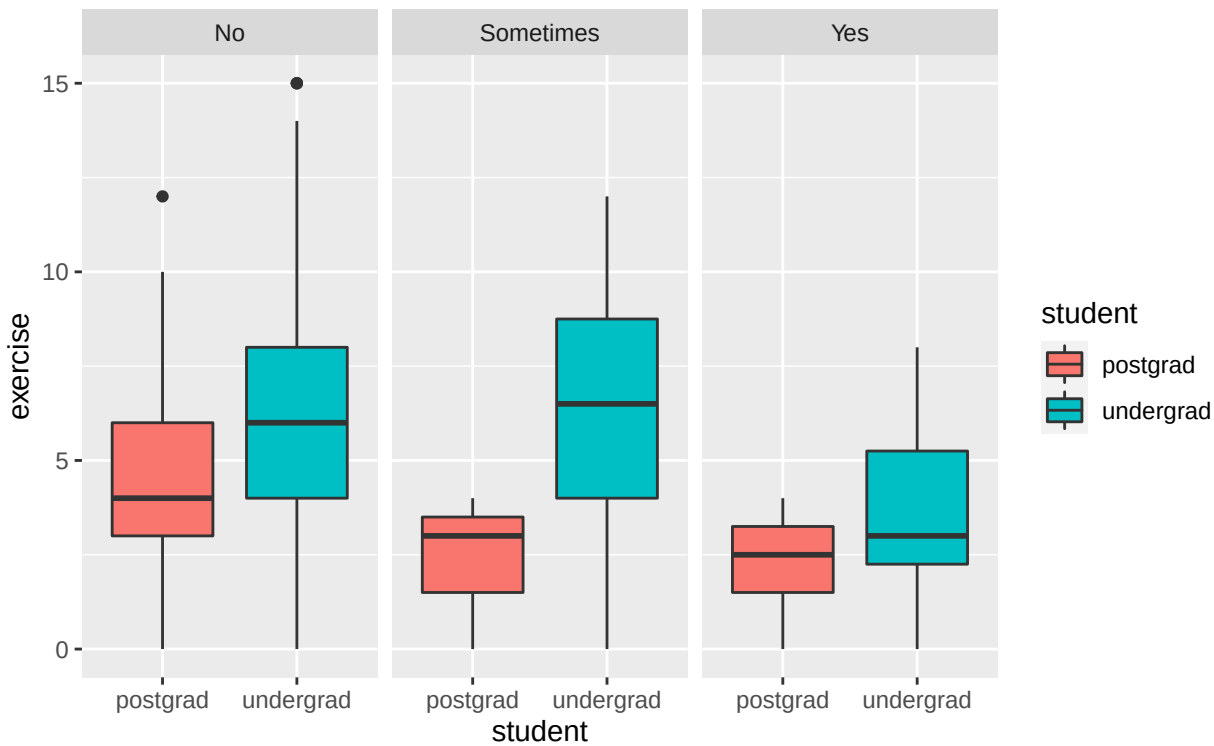
Question 3: [Total marks 30]

The HSE recommend we exercise for 2.5 hours per week. A University researcher set out to investigate whether average exercise differed between groups of students. They collected data from a sample of 244 students, asking how much they exercise per week, whether they were an undergraduate or postgraduate student and whether they smoked (no, sometimes or yes).

Descriptive statistics

```
# A tibble: 2 x 4
  student       n  mean    sd
  <chr>     <int> <dbl> <dbl>
1 postgrad     51  4.17  2.74
2 undergrad   193  6.29  3.31
```

```
# A tibble: 3 x 4
  smoke         n  mean    sd
  <fct>     <int> <dbl> <dbl>
1 No          189  6.00  3.29
2 Sometimes    41  5.99  3.38
3 Yes          14  3.36  2.47
```



a) An initial analysis using analysis of variance looked to see if there was any difference in average exercise between undergraduate and postgraduate students. Use the output below to test the hypothesis that there was no difference in mean exercise between the undergraduate and postgraduate students. Clearly specify your null and alternative hypotheses, calculate an appropriate F-statistic and use the tables at the end of this paper to perform a test at the 5% level. (4 marks)

```r
anova1 <- aov(exercise ~ student, data = surveydata)
summary(anova1)
```

```
             Df Sum Sq Mean Sq F value Pr(>F)
student       1  182.1  182.12       ?     ?
Residuals   242 2475.3   10.23
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
```

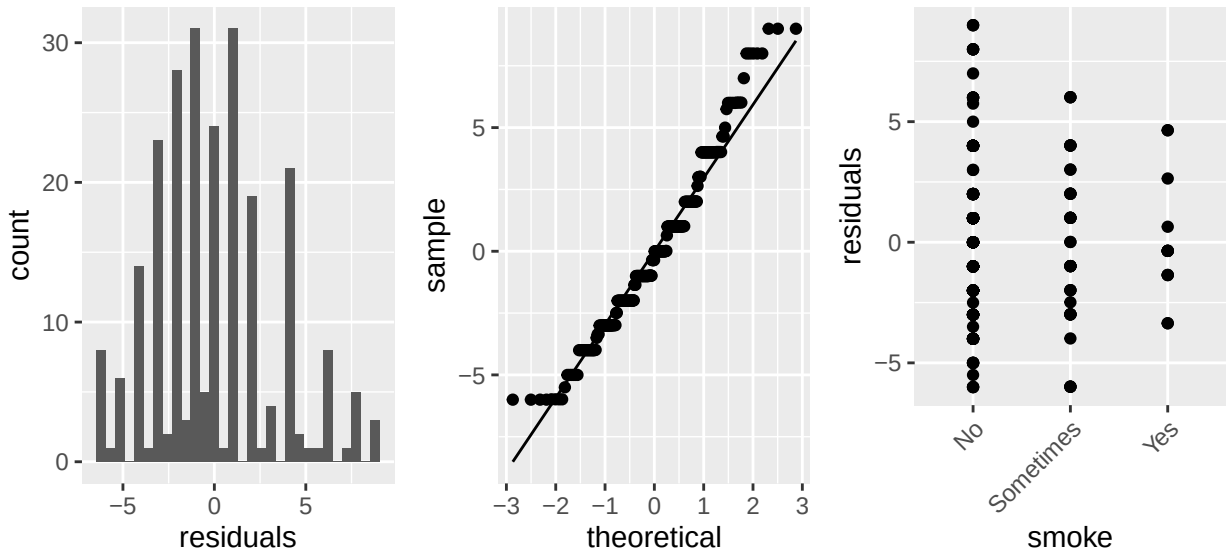
b) An alternative would have been to use a two-sample t-test; what would have been the value of the t-test statistic? (2 marks)

c) Next they looked to see if there was any difference due to smoking. Using the output below, test the hypothesis that there are no differences between the smoking groups. Again, clearly specify your null and alternative hypotheses, perform an F-test at the 5% level, and state your conclusions. (4 marks)

```r
anova2 <- aov(exercise ~ smoke, data = surveydata)
summary(anova2)
```

```
             Df Sum Sq Mean Sq F value Pr(>F)
smoke         2   92.1   46.05   4.327 0.0143 *
Residuals   241 2565.3   10.64
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

d) Comment on the validity of the assumptions for this ANOVA based on the residual plots below. If you have reservations on the validity, suggest what you might do to improve matters. (3 marks)



e) Following on from the above ANOVA why does it make sense to perform a multiple comparisons procedure? (1 mark)

f) Use the following output to perform multiple comparisons. Group the different smoking types accordingly and explain your decision using the relevant confidence intervals. (3 marks)

```
95 %-confidence intervals
 Method:  Difference of means assuming Normal distribution, allowing unequal variances


              estimate  lower   upper
Sometimes-No   -0.0135 -1.173  1.1458
Yes-No         -2.6442 -4.127 -1.1611
Yes-Sometimes  -2.6307 -4.354 -0.9076
```

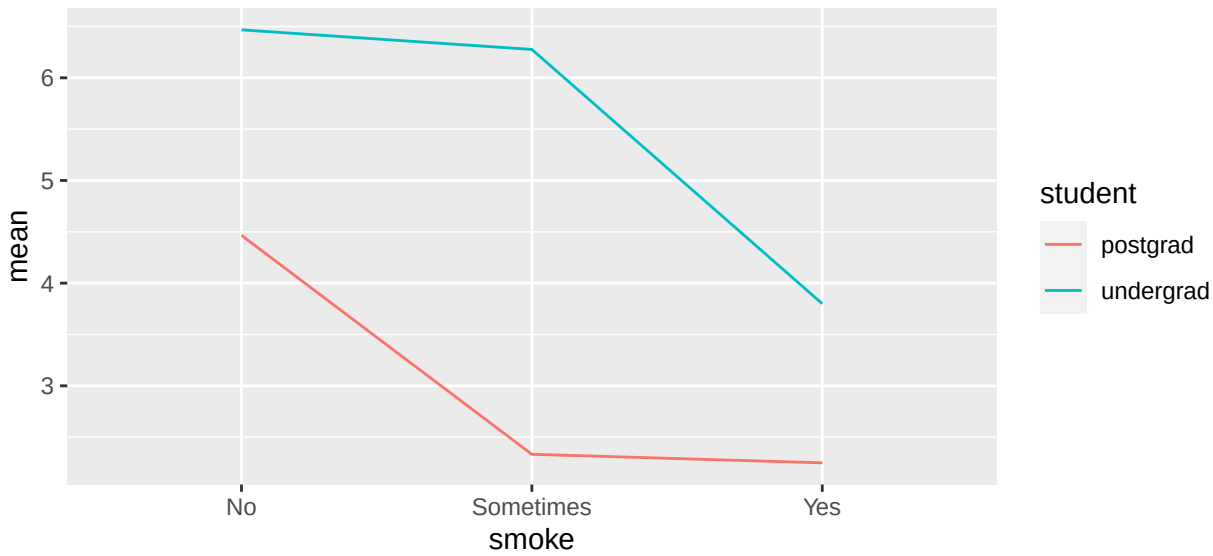
g) Define a suitable contrast to compare the population mean exercise between never smokers and ever smokers (i.e. Yes and Sometimes combined). Specify the null and alternative hypotheses being tested, obtain an estimate for the contrast statistic, calculate its standard error and the degrees of freedom for the test. Perform the test at the 5% significance level. (6 marks)

h) To consider both factors, `student` (i.e. Undergrad v Postgrad) and `smoke` (Yes, No, Sometimes), a two-way ANOVA was performed with the output given below. How does this test differ from the previous one way ANOVAs? What do you conclude? Relate your conclusions to your previous findings. (4 marks)

```
anova3 <- aov(exercise ~ student + smoke, data = surveydata)
summary(anova3)
```

```
             Df Sum Sq Mean Sq F value    Pr(>F)
student       1  182.1  182.12  18.284 0.0000275 ***
smoke         2   84.6   42.32   4.249    0.0154 *
Residuals   240 2390.6    9.96
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

i) Explain the interaction plot below. Does this suggest an interaction *might* be useful? Why? (1 mark)



j) The interaction model was fitted to the data, with output below. Interpret the result of the interaction model. (2 marks)

```
anova4 <- aov(exercise ~ student * smoke, data = surveydata)
summary(anova4)
```

```
               Df Sum Sq Mean Sq F value    Pr(>F)
student         1  182.1  182.12  18.213 0.0000285 ***
smoke           2   84.6   42.32   4.232    0.0156 *
student:smoke   2   10.6    5.32   0.532    0.5882
Residuals     238 2380.0   10.00
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Question 4: [Total marks 15]

Between February 25 and 27, 2004 there were six cases of Hepatitis A infection reported in Marshfield, Massachusetts. In addition, a case of Hepatitis A in a Plymouth resident, employed in Marshfield, was reported. Marshfield had reported just 1 case in all of 2002 and 0 cases in 2003. The increase in the number of reported cases during February in a confined geographic area was an indication of a possible outbreak of Hepatitis A infection.

In a situation like this, a case-control design is an efficient option to find the source of the outbreak. The investigators identified as many cases as possible (19 agreed to answer a questionnaire), and they selected a sample of 38 non-diseased people as a comparison group (the controls). Investigators then ascertained the prior exposures of subjects in each group, focusing on food establishments they had eaten at during the past two months.

Here's the results of whether cases and controls had eaten at Ron's Grill

	Ate at Ron's Grill	Did not eat at Ron's Grill	Total
Hepatitis Cases	18	1	19
Healthy Controls	7	31	38
Total	25	32	57

a) Compare Hepatitis A infection between those who ate at Ron's Grill and those who did not using the following measures and state what the relevant null and alternative hypotheses would be. You do not need to perform the test, simply calculate the test statistics:

(i) Equality of proportions;
(ii) Relative Risk;
(iii) Odds-ratio.

(6 marks)

b) For small studies, such as this, standard large-sample results based on the normal distribution can be unreliable. Fisher's exact test is designed to deal with this problem and calculates a randomization p-value based on the observed table and table(s) with the same marginal totals that show stronger evidence of a treatment effect. List the table(s) that are more extreme than the above table, **calculate the p-value** and provide your conclusion for this study. (9 marks)

Question 5: [Total marks 25]

Data are available from a vote in the U.S. Senate, which would prevent prescription drugs being imported from Canada and so lower drug prices. By voting with the bill senators are voting to keep drug prices high, by voting against the bill senators are voting to lower drug prices.

The primary research question here is to investigate the factors relating to the `Vote` of a senator with or against the bill (0=Against or 1=With). The possible predictors of this are the `Contributions` received by senators from the pharmaceutical industry over 6 years (in units of $1,000) and `Party` affiliation (D=Democrat or R=Republican).

```r
model1 <- glm(Vote ~ Contributions,
              data = CanadianDrugs, family = binomial(link = "logit"))
summary(model1)
```

```
Call:
glm(formula = Vote ~ Contributions, family = binomial(link = "logit"),
    data = CanadianDrugs)

Deviance Residuals:
    Min       1Q   Median       3Q      Max
-2.0597  -1.0449   0.4491   1.0408   1.5074

Coefficients:
               Estimate Std. Error z value Pr(>|z|)
(Intercept)   -0.785102   0.345663  -2.271 0.023129 *
Contributions  0.012081   0.003617   3.340 0.000838 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 129.25  on 93  degrees of freedom
Residual deviance: 114.06  on 92  degrees of freedom
AIC: 118.06

Number of Fisher Scoring iterations: 4
```

a) Write down the fitted model in both the logit and probability forms of the logistic regression model. (3 marks)
b) Interpret the estimated slope (i.e. `Estimate`) for `Contributions` and the odds ratio for `Contributions` once you have calculated it. (4 marks)
c) Is there a statistically significant relationship between voting for the bill and contributions from the pharmaceutical industry? State the null and alternative hypotheses,

associated p-value, and your conclusions. (4 marks)

d) Use the fitted model to predict the probability of voting against the bill for someone who received $80,000. (2 marks)

e) Calculate and interpret 95% confidence interval for the **odds ratio** of `Contributions`. (5 marks)

The previous analysis (`model1`) ignores possible differences between Republican and Democrat senators. To explore this two more models were fitted as shown below.

```r
model2 <- glm(Vote ~ Contributions + Party,
              data = CanadianDrugs, family = binomial(link = "logit"))
summary(model2)
```

```
Call:
glm(formula = Vote ~ Contributions + Party, family = binomial(link = "logit"),
    data = CanadianDrugs)

Deviance Residuals:
    Min       1Q   Median       3Q      Max
-2.0501  -0.5926   0.2533   0.8364   2.0383

Coefficients:
               Estimate Std. Error z value   Pr(>|z|)
(Intercept)   -2.319424   0.577736  -4.015 0.00005953 ***
Contributions  0.014688   0.004106   3.578   0.000347 ***
PartyR         2.509925   0.562860   4.459 0.00000823 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 129.246  on 93  degrees of freedom
Residual deviance:  88.922  on 91  degrees of freedom
AIC: 94.922

Number of Fisher Scoring iterations: 5
```

```r
model3 <- glm(Vote ~ Contributions * Party,
              data = CanadianDrugs, family = binomial(link = "logit"))
summary(model3)
```

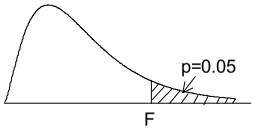
```
Call:
glm(formula = Vote ~ Contributions * Party, family = binomial(link = "logit"),
    data = CanadianDrugs)

Deviance Residuals:
    Min       1Q   Median       3Q      Max
-1.9588  -0.5668   0.3317   0.8095   2.0934

Coefficients:
                      Estimate Std. Error z value Pr(>|z|)
(Intercept)          -2.490648   0.666063  -3.739 0.000184 ***
Contributions         0.016324   0.005101   3.200 0.001373 **
PartyR                2.886891   0.864378   3.340 0.000838 ***
Contributions:PartyR -0.005077   0.008437  -0.602 0.547339
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 129.246  on 93  degrees of freedom
Residual deviance:  88.576  on 90  degrees of freedom
AIC: 96.576

Number of Fisher Scoring iterations: 5
```

f) Use the output (specifically the deviance values) from the three model fits to produce an analysis of deviance table for the effects of `Contributions`, `Party` and their interaction. What do you conclude from this table? Justify your conclusion with a formal significance test. (5 marks)

g) Use the interaction model estimates to calculate the odds ratio for a 1 unit (i.e. $1,000) increase in contributions for Democrat and Republican senators. (2 marks)

F distribution for p = 0.05

For various pairs of degress of freedom (v_1,v_2), the tabled entries represent the critical values of F above which the proportion p = 0.05 of the distribution falls. (Example, for df = 3,8 and F = 2.92 is exceeded by p = 0.05 of the population.

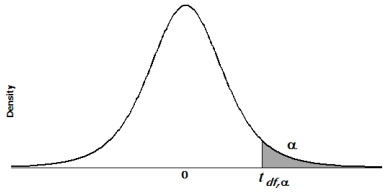


p = 0.05

	Degrees of freedom for the numerator v_1																	
v_2	1	2	3	4	5	6	7	8	9	10	12	15	20	30	40	60	120	∞
1	161.446	199.499	215.707	224.583	230.160	233.988	236.767	238.884	240.543	241.882	243.905	245.949	248.016	250.096	251.144	252.196	253.254	254.317
2	18.513	19.000	19.164	19.247	19.296	19.329	19.353	19.371	19.385	19.396	19.412	19.429	19.446	19.463	19.471	19.479	19.487	19.496
3	10.128	9.552	9.277	9.117	9.013	8.941	8.887	8.845	8.812	8.785	8.745	8.703	8.660	8.617	8.594	8.572	8.549	8.526
4	7.709	6.944	6.591	6.388	6.256	6.163	6.094	6.041	5.999	5.964	5.912	5.858	5.803	5.746	5.717	5.688	5.658	5.628
5	6.608	5.786	5.409	5.192	5.050	4.950	4.876	4.818	4.772	4.735	4.678	4.619	4.558	4.496	4.464	4.431	4.398	4.365
6	5.987	5.143	4.757	4.534	4.387	4.284	4.207	4.147	4.099	4.060	4.000	3.938	3.874	3.808	3.774	3.740	3.705	3.669
7	5.591	4.737	4.347	4.120	3.972	3.866	3.787	3.726	3.677	3.637	3.575	3.511	3.445	3.376	3.340	3.304	3.267	3.230
8	5.318	4.459	4.066	3.838	3.688	3.581	3.500	3.438	3.388	3.347	3.284	3.218	3.150	3.079	3.043	3.005	2.967	2.928
9	5.117	4.256	3.863	3.633	3.482	3.374	3.293	3.230	3.179	3.137	3.073	3.006	2.936	2.864	2.826	2.787	2.748	2.707
10	4.965	4.103	3.708	3.478	3.326	3.217	3.135	3.072	3.020	2.978	2.913	2.845	2.774	2.700	2.661	2.621	2.580	2.538
11	4.844	3.982	3.587	3.357	3.204	3.095	3.012	2.948	2.896	2.854	2.788	2.719	2.646	2.570	2.531	2.490	2.448	2.404
12	4.747	3.885	3.490	3.259	3.106	2.996	2.913	2.849	2.796	2.753	2.687	2.617	2.544	2.466	2.426	2.384	2.341	2.296
13	4.667	3.806	3.411	3.179	3.025	2.915	2.832	2.767	2.714	2.671	2.604	2.533	2.459	2.380	2.339	2.297	2.252	2.206
14	4.600	3.739	3.344	3.112	2.958	2.848	2.764	2.699	2.646	2.602	2.534	2.463	2.388	2.308	2.266	2.223	2.178	2.131
15	4.543	3.682	3.287	3.056	2.901	2.790	2.707	2.641	2.588	2.544	2.475	2.403	2.328	2.247	2.204	2.160	2.114	2.066
16	4.494	3.634	3.239	3.007	2.852	2.741	2.657	2.591	2.538	2.494	2.425	2.352	2.276	2.194	2.151	2.106	2.059	2.010
17	4.451	3.592	3.197	2.965	2.810	2.699	2.614	2.548	2.494	2.450	2.381	2.308	2.230	2.148	2.104	2.058	2.011	1.960
18	4.414	3.555	3.160	2.928	2.773	2.661	2.577	2.510	2.456	2.412	2.342	2.269	2.191	2.107	2.063	2.017	1.968	1.917
19	4.381	3.522	3.127	2.895	2.740	2.628	2.544	2.477	2.423	2.378	2.308	2.234	2.155	2.071	2.026	1.980	1.930	1.878
20	4.351	3.493	3.098	2.866	2.711	2.599	2.514	2.447	2.393	2.348	2.278	2.203	2.124	2.039	1.994	1.946	1.896	1.843
21	4.325	3.467	3.072	2.840	2.685	2.573	2.488	2.420	2.366	2.321	2.250	2.176	2.096	2.010	1.965	1.916	1.866	1.812
22	4.301	3.443	3.049	2.817	2.661	2.549	2.464	2.397	2.342	2.297	2.226	2.151	2.071	1.984	1.938	1.889	1.838	1.783
23	4.279	3.422	3.028	2.796	2.640	2.528	2.442	2.375	2.320	2.275	2.204	2.128	2.048	1.961	1.914	1.865	1.813	1.757
24	4.260	3.403	3.009	2.776	2.621	2.508	2.423	2.355	2.300	2.255	2.183	2.108	2.027	1.939	1.892	1.842	1.790	1.733
30	4.171	3.316	2.922	2.690	2.534	2.421	2.334	2.266	2.211	2.165	2.092	2.015	1.932	1.841	1.792	1.740	1.683	1.622
40	4.085	3.232	2.839	2.606	2.449	2.336	2.249	2.180	2.124	2.077	2.003	1.924	1.839	1.744	1.693	1.637	1.577	1.509
60	4.001	3.150	2.758	2.525	2.368	2.254	2.167	2.097	2.040	1.993	1.917	1.836	1.748	1.649	1.594	1.534	1.467	1.389
120	3.920	3.072	2.680	2.447	2.290	2.175	2.087	2.016	1.959	1.910	1.834	1.750	1.659	1.554	1.495	1.429	1.352	1.254
∞	3.841	2.996	2.605	2.372	2.214	2.099	2.010	1.938	1.880	1.831	1.752	1.666	1.571	1.459	1.394	1.318	1.221	1.000

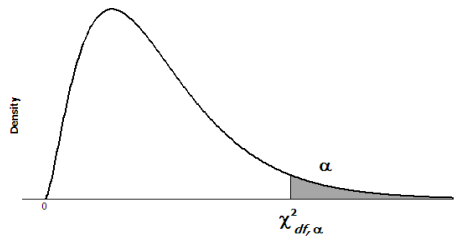
p.t.o.

Table: t distribution critical values



	Upper tail probability											
df	**0.25**	**0.20**	**0.15**	**0.10**	**0.05**	**0.025**	**0.02**	**0.01**	**0.005**	**0.0025**	**0.001**	**0.0005**
1	1.000	1.376	1.963	3.078	6.314	12.706	15.895	31.821	63.657	127.321	318.309	636.619
2	0.816	1.061	1.386	1.886	2.920	4.303	4.849	6.965	9.925	14.089	22.327	31.599
3	0.765	0.978	1.250	1.638	2.353	3.182	3.482	4.541	5.841	7.453	10.215	12.924
4	0.741	0.941	1.190	1.533	2.132	2.776	2.999	3.747	4.604	5.598	7.173	8.610
5	0.727	0.920	1.156	1.476	2.015	2.571	2.757	3.365	4.032	4.773	5.893	6.869
6	0.718	0.906	1.134	1.440	1.943	2.447	2.612	3.143	3.707	4.317	5.208	5.959
7	0.711	0.896	1.119	1.415	1.895	2.365	2.517	2.998	3.499	4.029	4.785	5.408
8	0.706	0.889	1.108	1.397	1.860	2.306	2.449	2.896	3.355	3.833	4.501	5.041
9	0.703	0.883	1.100	1.383	1.833	2.262	2.398	2.821	3.250	3.690	4.297	4.781
10	0.700	0.879	1.093	1.372	1.812	2.228	2.359	2.764	3.169	3.581	4.144	4.587
11	0.697	0.876	1.088	1.363	1.796	2.201	2.328	2.718	3.106	3.497	4.025	4.437
12	0.695	0.873	1.083	1.356	1.782	2.179	2.303	2.681	3.055	3.428	3.930	4.318
13	0.694	0.870	1.079	1.350	1.771	2.160	2.282	2.650	3.012	3.372	3.852	4.221
14	0.692	0.868	1.076	1.345	1.761	2.145	2.264	2.624	2.977	3.326	3.787	4.140
15	0.691	0.866	1.074	1.341	1.753	2.131	2.249	2.602	2.947	3.286	3.733	4.073
16	0.690	0.865	1.071	1.337	1.746	2.120	2.235	2.583	2.921	3.252	3.686	4.015
17	0.689	0.863	1.069	1.333	1.740	2.110	2.224	2.567	2.898	3.222	3.646	3.965
18	0.688	0.862	1.067	1.330	1.734	2.101	2.214	2.552	2.878	3.197	3.610	3.922
19	0.688	0.861	1.066	1.328	1.729	2.093	2.205	2.539	2.861	3.174	3.579	3.883
20	0.687	0.860	1.064	1.325	1.725	2.086	2.197	2.528	2.845	3.153	3.552	3.850
21	0.686	0.859	1.063	1.323	1.721	2.080	2.189	2.518	2.831	3.135	3.527	3.819
22	0.686	0.858	1.061	1.321	1.717	2.074	2.183	2.508	2.819	3.119	3.505	3.792
23	0.685	0.858	1.060	1.319	1.714	2.069	2.177	2.500	2.807	3.104	3.485	3.768
24	0.685	0.857	1.059	1.318	1.711	2.064	2.172	2.492	2.797	3.091	3.467	3.745
25	0.684	0.856	1.058	1.316	1.708	2.060	2.167	2.485	2.787	3.078	3.450	3.725
26	0.684	0.856	1.058	1.315	1.706	2.056	2.162	2.479	2.779	3.067	3.435	3.707
27	0.684	0.855	1.057	1.314	1.703	2.052	2.158	2.473	2.771	3.057	3.421	3.690
28	0.683	0.855	1.056	1.313	1.701	2.048	2.154	2.467	2.763	3.047	3.408	3.674
29	0.683	0.854	1.055	1.311	1.699	2.045	2.150	2.462	2.756	3.038	3.396	3.659
30	0.683	0.854	1.055	1.310	1.697	2.042	2.147	2.457	2.750	3.030	3.385	3.646
40	0.681	0.851	1.050	1.303	1.684	2.021	2.123	2.423	2.704	2.971	3.307	3.551
50	0.679	0.849	1.047	1.299	1.676	2.009	2.109	2.403	2.678	2.937	3.261	3.496
60	0.679	0.848	1.045	1.296	1.671	2.000	2.099	2.390	2.660	2.915	3.232	3.460
80	0.678	0.846	1.043	1.292	1.664	1.990	2.088	2.374	2.639	2.887	3.195	3.416
100	0.677	0.845	1.042	1.290	1.660	1.984	2.081	2.364	2.626	2.871	3.174	3.390
1000	0.675	0.842	1.037	1.282	1.646	1.962	2.056	2.330	2.581	2.813	3.098	3.300
Z*	0.674	0.842	1.036	1.282	1.645	1.960	2.054	2.326	2.576	2.807	3.090	3.291

Table: χ^2 distribution critical values



	Upper tail probability											
df	**0.25**	**0.20**	**0.15**	**0.10**	**0.05**	**0.025**	**0.02**	**0.01**	**0.005**	**0.0025**	**0.001**	**0.0005**
1	1.32	1.64	2.07	2.71	3.84	5.02	5.41	6.63	7.88	9.14	10.83	12.12
2	2.77	3.22	3.79	4.61	5.99	7.38	7.82	9.21	10.60	11.98	13.82	15.20
3	4.11	4.64	5.32	6.25	7.81	9.35	9.84	11.34	12.84	14.32	16.27	17.73
4	5.39	5.99	6.74	7.78	9.49	11.14	11.67	13.28	14.86	16.42	18.47	20.00
5	6.63	7.29	8.12	9.24	11.07	12.83	13.39	15.09	16.75	18.39	20.52	22.11
6	7.84	8.56	9.45	10.64	12.59	14.45	15.03	16.81	18.55	20.25	22.46	24.10
7	9.04	9.80	10.75	12.02	14.07	16.01	16.62	18.48	20.28	22.04	24.32	26.02
8	10.22	11.03	12.03	13.36	15.51	17.53	18.17	20.09	21.95	23.77	26.12	27.87
9	11.39	12.24	13.29	14.68	16.92	19.02	19.68	21.67	23.59	25.46	27.88	29.67
10	12.55	13.44	14.53	15.99	18.31	20.48	21.16	23.21	25.19	27.11	29.59	31.42
11	13.70	14.63	15.77	17.28	19.68	21.92	22.62	24.72	26.76	28.73	31.26	33.14
12	14.85	15.81	16.99	18.55	21.03	23.34	24.05	26.22	28.30	30.32	32.91	34.82
13	15.98	16.98	18.20	19.81	22.36	24.74	25.47	27.69	29.82	31.88	34.53	36.48
14	17.12	18.15	19.41	21.06	23.68	26.12	26.87	29.14	31.32	33.43	36.12	38.11
15	18.25	19.31	20.60	22.31	25.00	27.49	28.26	30.58	32.80	34.95	37.70	39.72
16	19.37	20.47	21.79	23.54	26.30	28.85	29.63	32.00	34.27	36.46	39.25	41.31
17	20.49	21.61	22.98	24.77	27.59	30.19	31.00	33.41	35.72	37.95	40.79	42.88
18	21.60	22.76	24.16	25.99	28.87	31.53	32.35	34.81	37.16	39.42	42.31	44.43
19	22.72	23.90	25.33	27.20	30.14	32.85	33.69	36.19	38.58	40.88	43.82	45.97
20	23.83	25.04	26.50	28.41	31.41	34.17	35.02	37.57	40.00	42.34	45.31	47.50
21	24.93	26.17	27.66	29.62	32.67	35.48	36.34	38.93	41.40	43.78	46.80	49.01
22	26.04	27.30	28.82	30.81	33.92	36.78	37.66	40.29	42.80	45.20	48.27	50.51
23	27.14	28.43	29.98	32.01	35.17	38.08	38.97	41.64	44.18	46.62	49.73	52.00
24	28.24	29.55	31.13	33.20	36.42	39.36	40.27	42.98	45.56	48.03	51.18	53.48
25	29.34	30.68	32.28	34.38	37.65	40.65	41.57	44.31	46.93	49.44	52.62	54.95
26	30.43	31.79	33.43	35.56	38.89	41.92	42.86	45.64	48.29	50.83	54.05	56.41
27	31.53	32.91	34.57	36.74	40.11	43.19	44.14	46.96	49.64	52.22	55.48	57.86
28	32.62	34.03	35.71	37.92	41.34	44.46	45.42	48.28	50.99	53.59	56.89	59.30
29	33.71	35.14	36.85	39.09	42.56	45.72	46.69	49.59	52.34	54.97	58.30	60.73
30	34.80	36.25	37.99	40.26	43.77	46.98	47.96	50.89	53.67	56.33	59.70	62.16
40	45.62	47.27	49.24	51.81	55.76	59.34	60.44	63.69	66.77	69.70	73.40	76.09
50	56.33	58.16	60.35	63.17	67.50	71.42	72.61	76.15	79.49	82.66	86.66	89.56
60	66.98	68.97	71.34	74.40	79.08	83.30	84.58	88.38	91.95	95.34	99.61	102.69
80	88.13	90.41	93.11	96.58	101.88	106.63	108.07	112.33	116.32	120.10	124.84	128.26
100	109.14	111.67	114.66	118.50	124.34	129.56	131.14	135.81	140.17	144.29	149.45	153.17